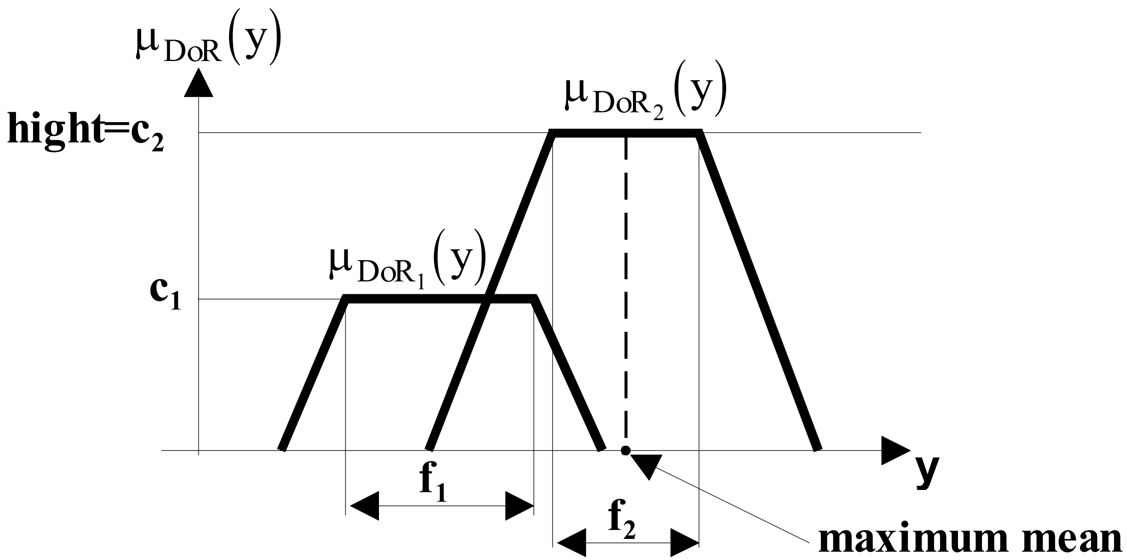
1.4. Defuzzification methods

Goal: Determine crisp y^* with the highest possibility based on.

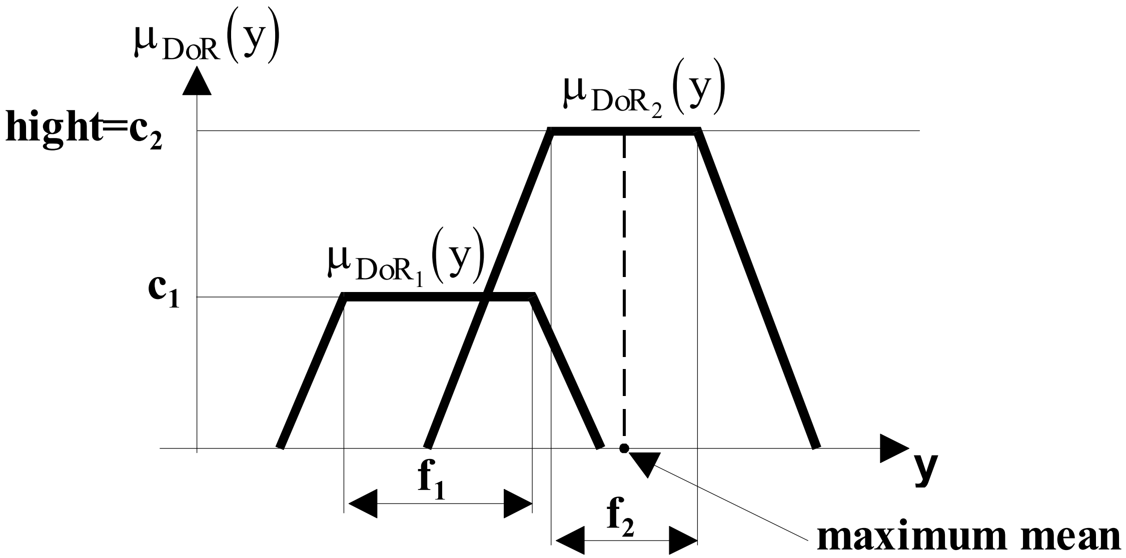
Center of gravity (COG) method



Continuous case: $y^*_{COA} = \dfrac{\int y\mu_{DoR}(y)dy}{\int \mu_{DoR}(y)dy}$, Discrete case: $y^*_{COA} = \dfrac{\sum_j y_j \mu_{DoR}(y_j)}{\sum_j \mu_{DoR}(y_j)}$.

- Neglects overlapping
- Cannot be computed from components $\mu_{DoR_i}(y) \Rightarrow$ Slow for real-time computation

Center of sums method (COS)



$$y^*_{COS} = \frac{\int y \sum_i \mu_{D\mathrm{o}R_i}(y)dy}{\int \sum_i \mu_{D\mathrm{o}R_i}(y)dy} = \frac{\sum_i \int y\mu_{D\mathrm{o}R_i}(y)dy}{\sum_i \int \mu_{D\mathrm{o}R_i}(y)dy}.$$

- The overlapped areas are considered with <u>multiplicity</u>
- The computation can be performed separately for each relation and the results can be summed after this

Bisector of area method (BOA)

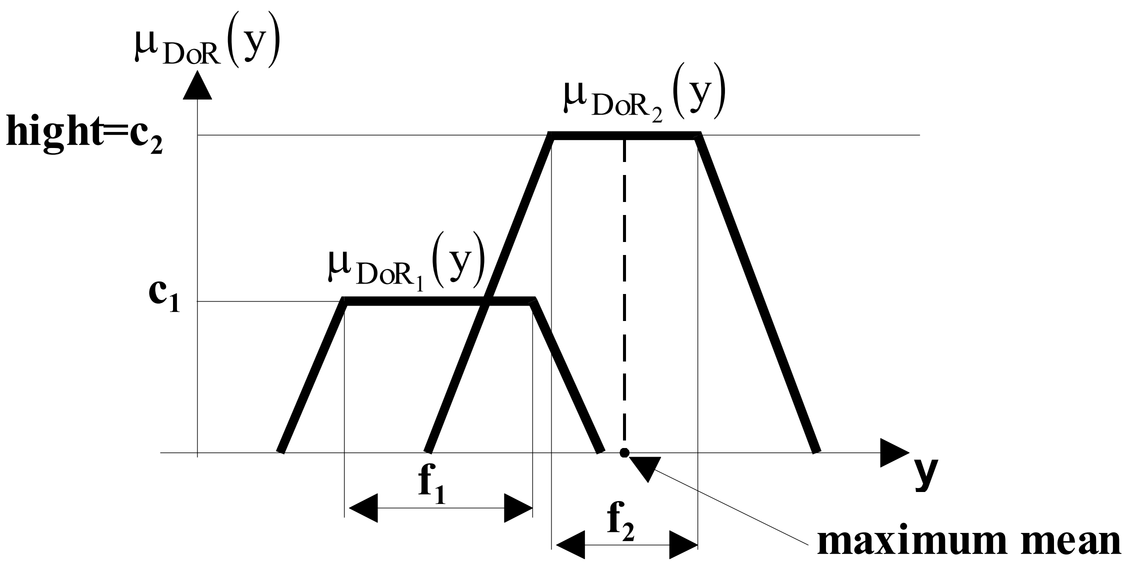
Denote a and b the minimum and maximum value of y, respectively, for which $\mu_{DoR}(y)$ is nonzero. Then the bisector of area (BOA) is y^*_{BOA} defined by:

$$\int_a^{y^*_{BOA}} \mu_{DoR}(y)\,dy = \int_{y^*_{BOA}}^{b} \mu_{DoR}(y)\,dy$$

Height method

Let the maximum of $\mu_{\mathrm{DoR}_i}(\mathrm{y})$ be denoted by c_i and let denote $\tilde{f}_i$ the place in y of the center of its plateau f_i (or alternatively $\tilde{f}_i := \arg\max Y^i(y)$). Then

$$y^*_{height} = \frac{\sum_i c_i \tilde{f}_i}{\sum_i c_i}.$$



Middle of Maxima (MOM)

It is called MOM (middle of maxima) method in literature, because it takes the middle of maxima (the middle of the plateau) of $\mu_{D \circ R}(y)$

$$hight = \max_{y} \mu_{D \circ R}(y)$$

$$y^*_{MOM} = \frac{\inf\{y : \mu_{D \circ R}(y) = \text{hight}\} + \sup\{y : \mu_{D \circ R}(y) = \text{hight}\}}{2}$$

Similarly we can choose the smallest of maximum ($y^*_{SOM} = \inf\{y : \mu_{D \circ R}(y) = \text{hight}\}$) or the largest of maximum ($y^*_{LOM} = \sup\{y : \mu_{D \circ R}(y) = \text{hight}\}$.

1.5. Sugeno-type fuzzy systems

The TSK- (Takagi-Sugeno-Kong) type or most often known as Sugeno-type fuzzy system unifies the principles of fuzzy and classical systems in such a manner which allows specifying deterministic consequence at the place of the consequence part of the various relations -

Benefit: It allows to use for example different deterministic algorithms in various operating points. The operating point is not crisp, but fuzzy, thus various operating points can exist simultaneously with different possibilities!

$$R_i: \textit{ if } x_i \textit{ is } X_1^i \textit{ and } \Lambda \textit{ and } x_N \textit{ is } X_N^i \textit{ then } y = f_i(x_1, \mathrm{K}, x_N), \quad i = 1, \mathrm{K}, n.$$

If $f_i = c_i$ constant $\Rightarrow$ Zero order Sugeno system

If $f_i = c_{i1}x_1 + \Lambda + c_{iN}x_N + c_{i0}$ linear function $\Rightarrow$ First order Sugeno system

TSK inference and defuzzification algorithm:

1) Inference:

$$x_1^*, x_2^*, \mathrm{K}, x_N^* \text{ Crisp inputs}$$

$$\forall R_i \; \alpha \; \left(\tau_i, y_i^*\right)$$

$$\tau_i = \mu_{X_1^i}\left(x_1^*\right) \wedge \Lambda \wedge \mu_{X_N^i}\left(x_N^*\right)$$

$$y_i^* = f_i\left(x_1^*, \Lambda, x_N^*\right)$$

2) Defuzzification:

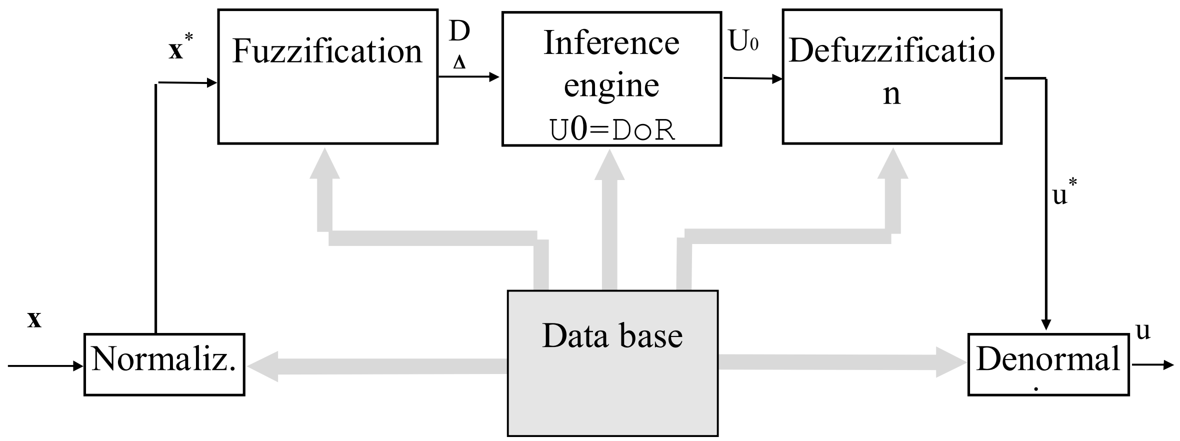
$$y_{TSK}^* = \frac{\sum_i \tau_i y_i^*}{\sum_i \tau_i}.$$

Application: Widespread (helicopter, vehicle control, robotics)

1.6. Fuzzy Logic Controllers- FLCs

Controller input: control error, measured or observed state variables.
Controller output: The input of controlled process

1.6.1. The structure of the fuzzy logic controller



x vector:	The signals of process (reference signal, senspors, etc)
Normalization:	The purpose of normalization is to provide that the membership function transforms the process signal x_i to (in most case) normalized domain: $x_i^* = N_{xi} x_i$. The normalization in the system corresponds to the gain factor.
Fuzzification:	Assign a fuzzy set A_i to the crisp or noisy normalized variable x_i^*. The aggregation of these formulates the input data D of the inference engine.
Inference engine:	The inference engine performs the inference based on the input data and the rules describing the behavior of the controller (rule base), which results in a fuzzy set U_0 (and its membership function $\mu_{U_0}(u)$).
Defuzzification:	creates normalized crisp controller output u^* based on U_0
Denormalization:	$u = N_u u^*$ gain factor for actuator.
Data base:	Normalization parameters, Rules of fuzzification and parameters of membership function, Rule base of inference, defuzzification rules, parameters of denormalization.

Nonlinear controller! (max, min nonlinear function in the inference algorithm)

Strategies for normalization:

1. Formulate the rules of the FLC in the natural domain of the variables x_i, u without the application of normalization and denormalization (in this case we must take into consideration the real technical limits of the variables during the elaboration of the rules).
2. formulate the rules of the FLC in the standardized domain of the variables x_i^*, u^* (which may be $[-1,1]$ or $[0,1]$) and apply normalization and denormalization gains.

   (In this case we do not need to take into consideration the technical limits of the variables during the elaboration of the rules, but we must find appropriate gain factors for the normalization and the denormalization.).

1.6.2. PID-like fuzzy logic controller

Reference signal: y_d
Process output: y
Error: $e = y_d - y$

Classic PID controller: $u = K_P e + K_I \int e dt + K_D \frac{de}{dt}$, i.e

$$u_k = K_P e_k + K_I T \Sigma e_k + \frac{K_D}{T} \Delta e_k$$

Introducing $k_p = K_p$, $k_I = K_I T$, $k_D = K_D / T$, the output of PID cpntroller:

$$u_k = k_p e_k + k_I \Sigma e_k + k_D \Delta e_k$$
$$u_{k-1} = k_p e_{k-1} + k_I \Sigma e_{k-1} + k_D \Delta e_{k-1}$$

Considering $\Sigma e_k - \Sigma e_{k-1} = e_k$, $\Delta e_k - \Delta e_{k-1} = e_k - 2e_{k-1} + e_{k-2}$, the increment $\Delta u_k = u_k - u_{k-1}$ of controller output is:

$$\Delta u_k = k_p(e_k - e_{k-1}) + k_I e_k + k_D(e_k - 2e_{k-1} + e_{k-2}) = (k_p + k_I + k_D)e_k + (-k_p - 2k_D)e_{k-1} + k_D e_{k-2}$$

$$\Delta u_k = k_0 e_k + k_1 e_{k-1} + k_2 e_{k-2}$$

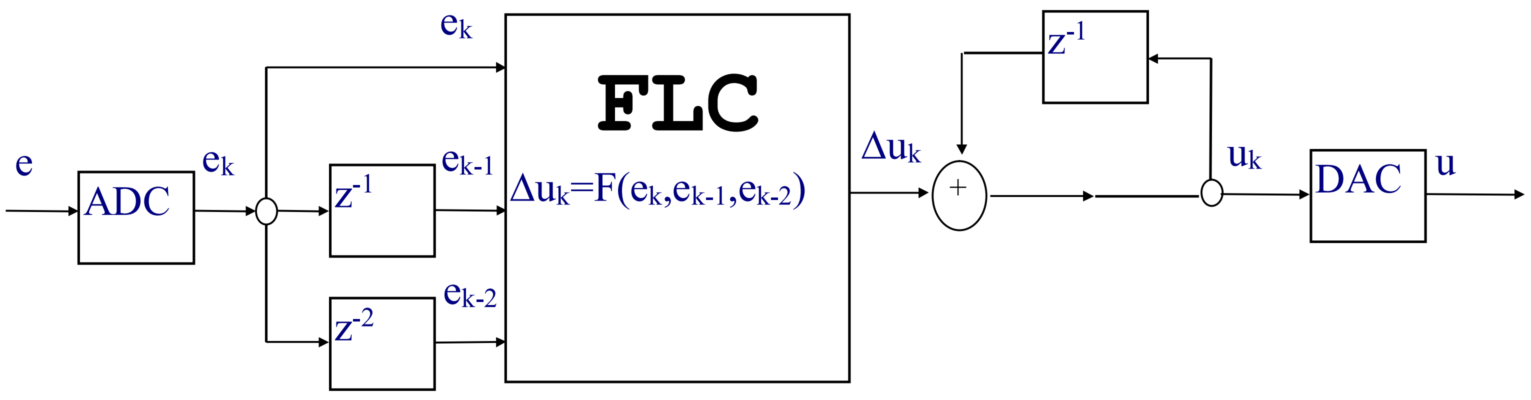
The implication rule of a PID-like fuzzy logic controller (PID-like FLC) has the form::

$$R_i : if\ \ e_k\ \ is\ \ E_k^i\ \ and\ \ e_{k-1}\ \ is\ \ E_{k-1}^i\ \ and\ \ e_{k-2}\ \ is\ \ E_{k-2}^i\ \ then\ \ \Delta u_k\ \ is\ \ \Delta U_k^i$$

Ninlinear controller (using min, max, etc):

$$\Delta u_k = F(e_k, e_{k-1}, e_{k-2})$$
$$u_k = u_{k-1} + \Delta u_k$$

Schematics (PID-like fuzzy controller)



- The number of possible rules are large because of the three variables $\Rightarrow$
- hard to construct the controller rules $\Rightarrow$
- Use PD or PI-like controller instead

1.6.3. PD-like controller

The algorithm: $u_k = k_p e_k + k_D \Delta e_k$

Normalization:

$e_{\max} = y_{d,\max} - y_{\min}$

$e_{\min} = y_{d,\min} - y_{\max} = -e_{\max}$

$\Delta e_{\max} = e_{\max} - e_{\min}$

$\Delta e_{\min} = e_{\min} - e_{\max} = -\Delta e_{\max}$

$e \in [-a_e, a_e]$

$e^* \in [-a_e^*, a_e^*]$

$\Delta e \in [-a_{\Delta e}, a_{\Delta e}]$

$\Delta e^* \in [-a_{\Delta e}^*, a_{\Delta e}^*]$

$u \in [-a_u, a_u]$

$u^* \in [-a_u^*, a_u^*]$

$a_e^* = N_e a_e$

$a_{\Delta e}^* = N_{\Delta e} a_{\Delta e}$

$$\boxed{\begin{aligned} N_e &= a_e^*/a_e \\ N_{\Delta e} &= a_{\Delta e}^*/a_{\Delta e} \end{aligned}}$$

Attention! Since u should be denormalized, the parameter of normalized domain is in the denominator (not in the numerator):

$a_u = N_u a_u^*$

$$\boxed{N_u = a_u/a_u^*}$$

Consequences of the bad choces of normalization parameters

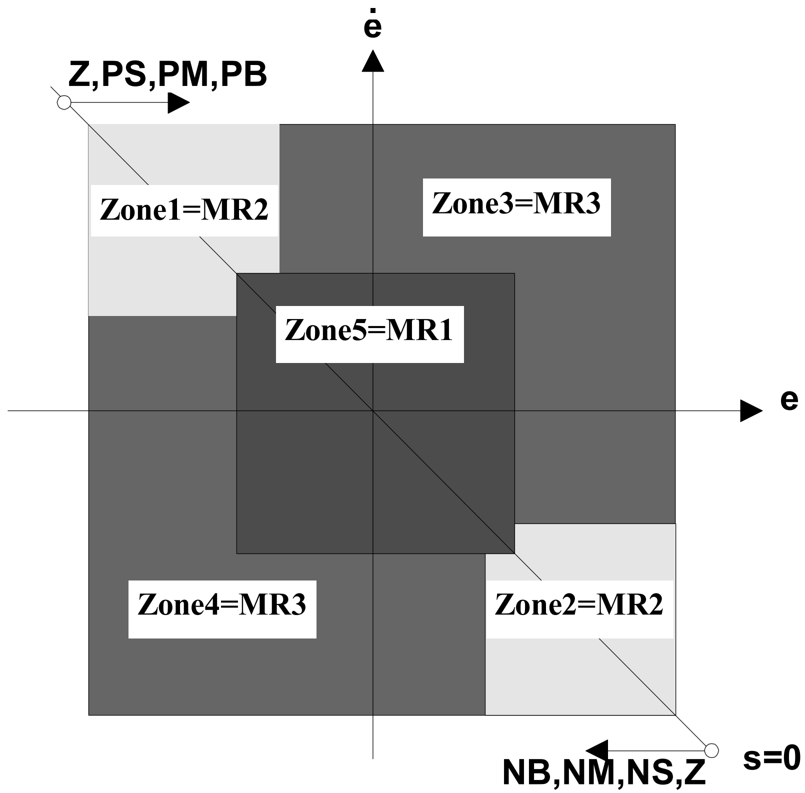
- the domain of operation is mapped on only to a small part of the normalized domain (sensitivity problem) **or**
- before reaching the boundary of the domain of operation the normalized output knocks against the boundary of the normalized domain (saturation problem)

Determination of the rule base:

Three meta rules by MacVicar-Whelan:

MR1) If the error e_k and the change of error Δe_k are zero, then maintain the present control setting.
MR2) If the error e_k is tending to zero at a satisfactory rate, then maintain the present control setting
MR3) If the error e_k is not self-correcting, then control action Δu_k is not zero and depends on the sign and magnitude of e_k and Δe_k.

Let the normalized gain factors be $K_{PN}, K_{DN} = 1$. Let us use the notation $s = u$ and consider the "switching curve" $s = 0$ in the $(\dot{e}, e)$ state space: $s = e + \dot{e} = 0 \Rightarrow \dot{e} = -e$.



- **Zone 5:** e and $\dot{e} \sim \Delta e$ are small, therefore MR1 becomes operative
- **Zone 1** and **Zone 2** the error converges to zero with a satisfactory rate, therefore MR2 is the operating rule.
- **Zone 3** és **Zone 4** error is not self-correcting, therefore MR3 is the operating rule.

When choosing the rules

- Strive for symmetry
- Because of the interpretation of negative feedback and $e = y_d - y$, we have to assure $u > 0$ above the switching curve $s = 0$, and $u < 0$ below the switching curve, respectively. Along the switching curve, e and $\dot{e} \sim \Delta e$ have equal order of magnitudes and their signs are the opposite, therefore $u = e + \dot{e} = 0$ can be chosen, which corresponds to the fuzzy value Z.
- Since in the top-right corner of the figure $e = \dot{e} \sim \Delta e$ is not self-correcting ($\dot{e} \sim \Delta e$ further increases the error), therefore, advancing from the left-top corner of the figure toward the right-top corner, gradually Z,PS,PM,PB controller change should be prescribed
- Similarly, advancing from the bottom-right corner toward the bottom-left corner, gradually Z,NS,NM,NB fuzzy values should be prescribed
- The fuzzy relations

$$R_i : if \;\; e \;\, is \;\, E^i \;\; and \;\; \Delta e \;\, is \;\, \Delta E^i \;\; then \;\, u \;\, is \;\, U^i$$

can be defined in a table by giving the value U^i at the intersection of column E^i and row ΔE^i:

Δe	PB	Z	PS	PM	PM	PB	PB	PB
	PM	NS	Z	PS	PM	PM	PB	PB
	PS	NM	NS	Z	PS	PM	PM	PB
	Z	NM	NM	NS	Z	PS	PM	PM
	NS	NB	NM	NM	NS	Z	PS	PM
	NM	NB	NB	NM	NM	NS	Z	PS
	NB	NB	NB	NB	NM	NM	NS	Z
		NB	NM	NS	Z	PS	PM	PB
		e						

Introducing Fuzzy Toolbox